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CYBER

THREAT INTELLIGENCE • MALWARE ANALYSIS • DETECTION ENGINEERING

Aviation Weather and Runway Data Integrity Cyber Risk Daily Review Update

Threat Report

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Author	Lost Boys Cyber
Report Type	Threat Report

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1. Executive Summary

Aviation operators rely on trusted weather, runway, NOTAM, and dispatch data. Although the daily source set did not identify a new destructive campaign against these systems, data-integrity risk remains a priority because disruption or manipulation can create operational delays and safety-sensitive decision pressure.

This daily review prioritizes practical defender action over attribution certainty. Source depth was expanded to the last seven days where needed to preserve high-confidence, multi-source reporting rather than fabricate same-day claims.

2. Intelligence Requirements

Question	Current Answer	Confidence
What changed for defenders today?	Operational-data dependency remains an aviation-specific cyber risk even without a fresh named intrusion.	Medium
Who is most exposed?	Airports, airlines, dispatch operations, weather-data consumers, and data integrators.	Medium
What immediate action should analysts take?	Validate data-source integrity monitoring, access controls, and manual fallback procedures.	High

3. Source Review & Confidence

Source	Contribution	Confidence
NOAA Aviation Weather Center	Authoritative aviation weather service dependency context	High
EASA aviation cybersecurity threat landscape project	Sector cybersecurity context	High
Military Aerospace connected aviation cyber challenges	Connected aviation risk context	Medium

4. Threat Activity Overview

The likely exposure path is compromise of supporting IT, supplier APIs, credentials, or data pipelines rather than direct aircraft compromise. Defenders should treat data integrity as part of cyber resilience and incident response planning.

5. Affected Sectors and Exposure

Sector / Environment	Exposure	Analyst Priority
Enterprise identity and endpoint estates	Credential abuse, script execution, and malware/ransomware staging.	Review high-risk sign-ins, new persistence, and anomalous script activity.
Cloud and SaaS administration	Token theft, OAuth abuse, risky automation, and data exposure.	Audit consent grants, service principals, and admin actions.
Managed service / supply chain	Shared access paths amplify impact.	Validate supplier controls and inbound trust

		relationships.

6. Aviation and Aerospace Sector Impact

Segment	Operational Relevance	Exposure Path	Defender Action
Airports, flight operations centers, dispatch, ATC support functions, and aviation weather data consumers	Weather, runway, NOTAM, and operational data integrity influence safe and efficient aviation decisions.	Identity, supplier access, exposed services, SaaS, OT/IT integration, or satellite/ground-system dependency.	Validate segmentation, vendor access, MFA posture, backups, and incident communications.
Third-party ecosystem	Aviation depends on MRO, airport, reservation, weather, satellite, and defense supply-chain partners.	Compromised managed service, shared platform, or supplier credential.	Require supplier attestations for patching, logging, and notification SLAs.

7. Tactics, Techniques, and Procedures

Tactic / Phase	Observed Technique	Evidence
Initial Access	Phishing, exposed service exploitation, or supplier compromise	Source reporting describes active threat activity or sector-relevant risk.
Execution	Command/script execution and automation abuse	Daily detection focus includes process and cloud-control-plane telemetry.
Persistence / Defense Evasion	Token/OAuth misuse, service changes, tooling overlap	Hunt for abnormal long-lived access and administrative changes.
Impact	Data theft, ransomware/extortion, or operational disruption	Reported campaigns and sector risk concentrate on business disruption.

8. Infrastructure and Indicators

Indicator / Infrastructure	Type	Operational Use
Public sources did not provide stable atomic IOCs for all environments	Limitation	Use behavior-led detections and enrich with local/vetted vendor IOCs.
aviation weather data	Keyword / analytic pivot	Use in triage notes, EDR search, and CTI tagging; not an IOC by itself.

9. Detection Engineering

9.1. Detection Summary

Signal	Telemetry Source	Analytic Goal
Suspicious script or command execution after lure/access event	EDR process telemetry	Identify payload staging, reconnaissance, or automation abuse.
Unusual outbound destination or direct-to-IP access	Network/EDR DNS and connection telemetry	Surface malware C2, staging, or infrastructure bypassing DNS controls.
New OAuth grant, token use, or admin consent anomaly	Cloud identity logs	Detect identity-centric intrusion and persistence.

9.2. Sigma / Detection Content

```
title: Daily Threat Activity Signal - aviation weather data
id: daily-20260814-daily-threat-activity-signal-aviation-weather-da
status: experimental
description: Analyst-facing daily-review detection seed for Daily Threat Activity Signal - aviation
weather data.
references:
  - https://github.com/lost0x01/lost-boys-cyber-analyst-kit
author: Lost Boys Cyber
date: 2026/08/14
logsource:
  category: process_creation
detection:
  selection_keywords:
    Keyword1: "aviation weather data"
    Keyword2: "ransomware"
    Keyword3: "token"
  condition: selection_keywords
falsepositives:
  - Administrative scripts or vulnerability-scanner probes using the same product strings.
level: medium
```

9.3. Hunt Query

```
let lookback = 14d;
let terms = dynamic(["aviation weather data", "ransomware", "oauth", "token"]);
DeviceNetworkEvents
| where Timestamp > ago(lookback)
| where RemoteUrl has_any (terms) or InitiatingProcessCommandLine has_any (terms)
| summarize FirstSeen=min(Timestamp), LastSeen=max(Timestamp), Hosts=dcount(DeviceName),
Samples=make_set(InitiatingProcessCommandLine, 5) by RemoteUrl, RemoteIP, InitiatingProcessFileName
| order by LastSeen desc
```

10. Threat Hunting

10.1. Hunt Hypothesis

Relevant activity will present as behavior chains rather than a single static IOC: lure or exploit trigger, script/process execution, identity or token changes, outbound staging, and data-access anomalies.

10.2. Hunt Query

```
let lookback = 14d;
let pivots = dynamic(["aviation weather data", "oauth", "token", "ransomware", "extortion"]);
DeviceProcessEvents
| where Timestamp > ago(lookback)
| where ProcessCommandLine has_any (pivots) or InitiatingProcessCommandLine has_any (pivots)
| summarize FirstSeen=min(Timestamp), LastSeen=max(Timestamp), Hosts=dcount(DeviceName),
Examples=make_set(ProcessCommandLine, 5) by FileName, InitiatingProcessFileName
| order by LastSeen desc
```

11. Risk Assessment

Risk	Likelihood	Impact	Notes
Identity-led intrusion	Medium	High	Current reporting continues to show credential/token misuse as a durable access path.
Ransomware/extortion	Medium	High	Impact depends on segmentation, backup readiness, and supplier blast radius.

Operational disruption	Medium	High	Highest for aviation/aerospace, healthcare, manufacturing, and managed-service dependencies.

12. Recommended Actions

Priority	Action
P0	Review whether the organization uses affected products/services or matches the reported exposure pattern.
P1	Run the included hunts, tune with local IOCs, and validate alerts with identity/network context.
P1	Confirm MFA, conditional access, vendor access, backups, and incident communications for affected business owners.
P2	Add source URLs and report metadata to CTI tracking/OpenCTI for recurring review.

13. References

- [1] NOAA Aviation Weather Center: https://aviationweather.gov/
- [2] EASA aviation cybersecurity threat landscape project: https://www.easa.europa.eu/en/research-projects/cyber
- [3] Military Aerospace connected aviation cyber challenges: https://www.militaryaerospace.com/commercial-aerospace/news/55385521/cybersecurity-challenges-grow-across-connected-aircraft-and-satellite-networks